Rhutam Mahajan

College Park, MD

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Education

University of Maryland, College Park **Aug 2025 – May 2027**
M.S. in Data Science | GPA: 3.95/4.0 *College Park, MD*

Savitribai Phule Pune University **Dec 2021 – Jun 2025**
B.E. in Computer Engineering, Honors in Data Science | GPA: 8.5/10 *Pune, India*

Technical Skills

Languages & Core Libraries: Python, SQL, Pandas, NumPy, SciPy, scikit-learn
Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, Hugging Face Transformers, computer vision, NLP, few-shot and multimodal learning, time series forecasting
Generative AI & LLMs: OpenAI Agents SDK, Model Context Protocol (FastMCP), LangChain, LangGraph, ChromaDB, RAG, tool calling
Big Data, Cloud & Tools: Apache Spark, PySpark, Spark MLlib, Neo4j, MySQL, AWS (ECR, S3, App Runner), Docker, FastAPI, Git, Tableau, Power BI

Experience

InternPe **Feb 2024 – Mar 2024**
AI/ML Intern *Remote*

- Developed four ML projects spanning classification, regression, and neural networks; deployed three as live Streamlit apps.
- Fixed train/test leakage in an IPL win predictor by splitting on matches, not deliveries, for a 73% test accuracy.
- Engineered a diabetes screening model to 0.81 ROC AUC; handled invalid zero readings and raised recall from 0.50 to 0.69.
- Reached a cross validated R^2 of 0.64 on used car prices; a Keras net matched a logistic baseline on cancer data at 0.99 AUC.

Gamaka AI **Dec 2023 – Mar 2024**
Junior Data Scientist *Pune, India*

- Trained classification, regression, clustering, and time series models across 11 projects, reaching 97.1% on MNIST digits.
- Compared KNN, SVM, Naive Bayes, and LSTM classifiers on SMS spam; the LSTM led with 99.3% test accuracy.
- Designed normalized MySQL schemas of up to 12 tables and delivered two Power BI dashboards on sales and banking data.
- Wrote SQL queries with self joins over a financial datasets to report revenue, profit, and EPS growth.

Projects

Agentic AI Systems with MCP, RAG, and Evaluator Loops **Apr 2026**

- Architected async multi-agent workflows across research, tools, memory, retrieval, and evaluation over a custom FastMCP server.
- Created a RAG chatbot with ChromaDB vector search, OpenAI generation, and Gemini-based evaluation.
- Orchestrated tool-calling loops so agents could plan a step, call a function, and revise the answer from evaluator feedback before replying.
- Refined the prompts and retrieval from the logged evaluator scores, which noticeably reduced hallucinated responses.

Spark Readmission Pipeline & Graph Data Engineering **Apr 2026**

- Constructed a PySpark pipeline on 12,465 records; generated 16 features via Spark ML and mapped relationships in Neo4j.
- Wrangled raw admission records in Spark and resolved missing fields and skewed distributions before any modeling.
- Queried the Neo4j graph with Cypher to surface patient and provider relationships that the flat features missed.
- Trained a Spark MLlib Random Forest: 60.57% accuracy, 58.29% F1, and 0.636 AUC on held-out data.

Multimodal Medical Triage **Dec 2025 – Mar 2026**

- Fused ResNet-18 lesion-image features with clinical text in a PyTorch model: 80.13% accuracy, 77.32% macro F1.
- Balanced the image and text branches during fusion so one modality would not drown out the other.
- Added a confidence threshold that sent low-certainty cases to manual review rather than a forced prediction.
- Shipped a FastAPI + React/TypeScript app, Dockerized and deployed to AWS ECR, App Runner, and S3 via GitHub Actions.

Plant Leaf Disease Multilevel Classification with Few-Shot Learning **Aug 2024 – May 2025**

- Devised a two-level few-shot classifier on EfficientNet-B0 with Prototypical Networks that mapped 38 classes to 12.
- Sampled episodic support and query sets so the prototypical network learned in a genuine few-shot setting.
- Trained EfficientProtoNet on 240 images: 97.77% accuracy, 95.83% macro F1 on 13,135 held-out images.
- Published the work in IJRAR, an international peer-reviewed journal, Vol. 12, Issue 2, May 2025.